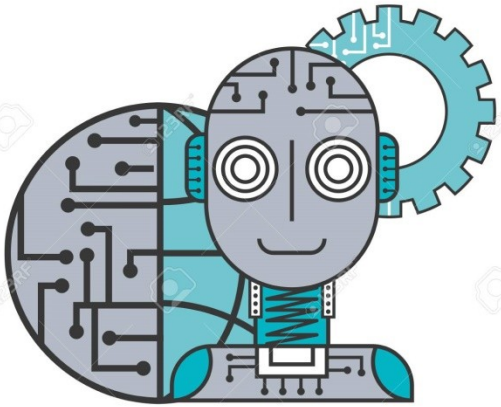
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Module 6CM521: Frontiers in AI and Data Science



EDA

Ioannis Tsioulis (MC)

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Slides acknowledgements:
Dr Hong Qing Yu

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Declaration

This lecture notes are organised based on Book *Doing Data Science, Chapter 2, EDA section*

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EDA is an attitude

“Exploratory data analysis” is an attitude, a state of flexibility, a willingness to look for those things that we believe are not there, as well as those we believe to be there.

— John Tukey

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What is EDA?

Exploratory Data Analysis (EDA) is an approach/philosophy for data analysis that employs a variety of techniques (mostly graphical) to

- maximize insight into a data set;
- uncover underlying structure;
- extract important variables;
- detect outliers and anomalies;
- test underlying assumptions;
- develop parsimonious models; and
- determine optimal factor settings.

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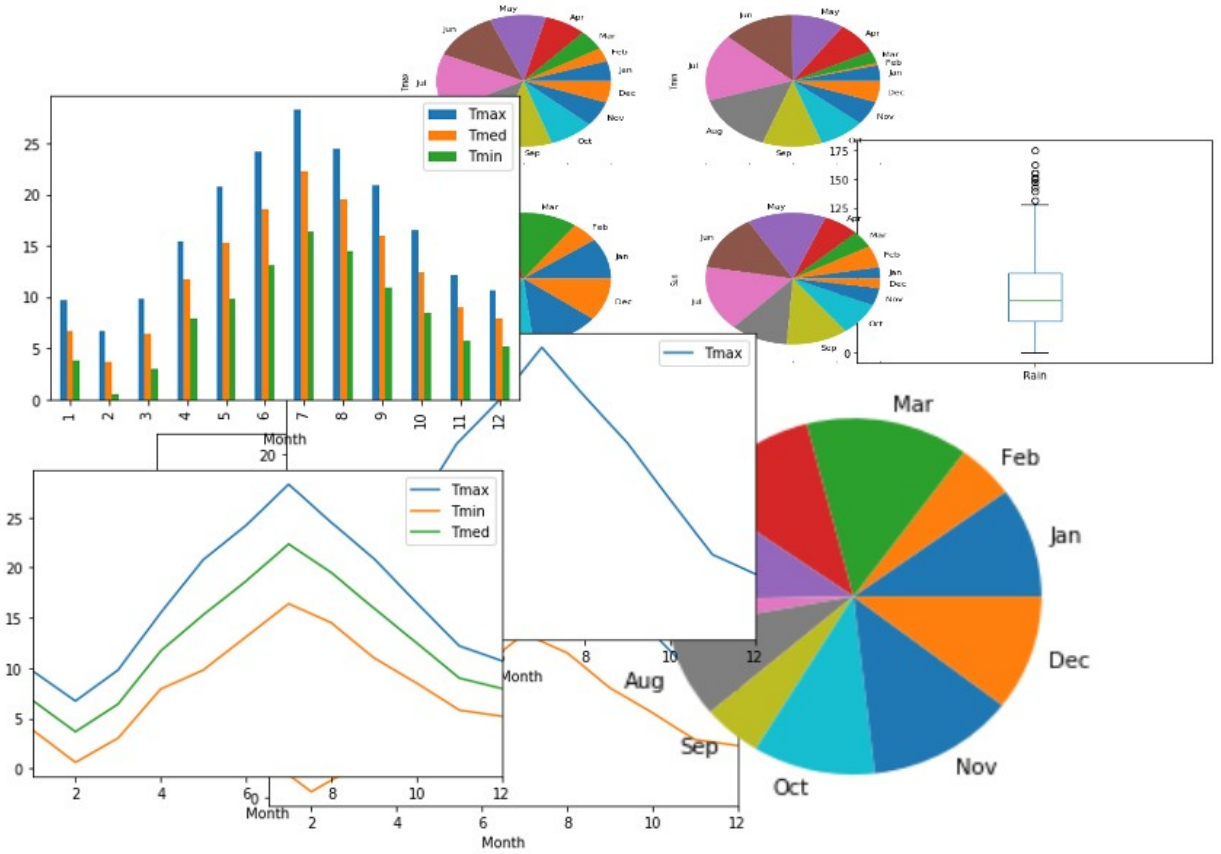
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Tools of EDA

- The basic tools of EDA are plots, graphs and summary statistics (e.g. data distributions).



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EDA Mindset

But as much as EDA is a set of tools, it's also a mindset. And that mindset is about your relationship with the data.



EDA happens between you and the data and isn't about proving anything to anyone else yet.

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Philosophy of EDA

Long before worrying about how to convince others, you first have to understand what's happening yourself.

— Andrew Gelman

- Even with very large Google-scale data, they did EDA.
- In the context of data in an Internet/engineering company, EDA is done for some of the same reasons it's done with smaller datasets, but there are additional reasons to do it with data that has been generated from logs.

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Philosophy of EDA (Cont.)

- There are important reasons anyone working with data should do EDA. Namely, to gain intuition about the data; to make comparisons between distributions; for sanity checking (making sure the data is on the scale you expect, in the format you thought it should be); to find out where data is missing or if there are outliers; and to summarize the data.
- In the context of data generated from logs, EDA also helps with debugging the logging process. For example, “patterns” you find in the data could actually be something wrong in the logging process that needs to be fixed. If you never go to the trouble of debugging, you’ll continue to think your patterns are real. The engineers we’ve worked with are always grateful for help in this area.
- In the end, EDA helps you make sure the product is performing as intended.

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Who are they?

John Tukey, a mathematician at Bell Labs, developed exploratory data analysis in contrast to confirmatory data analysis, which concerns itself with modelling and hypotheses.

Exploratory Data Analysis by John Tukey (Pearson)

"Exploratory Data Analysis for Complex Models" by Andrew Gelman (American Statistical Association, MIT)

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6 steps to do your EDA

1. Import Dataset & Headers
2. Organizing your data (Correct Data Formats)

3.1. Identify Missing Data
3.2 Replace Missing Data
3.3 Evaluate Missing Data
3.4 Dealing with Missing Data

Data pre-processing

4. Indicator variable
5. Data standardization
6. Data Normalization

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What is data?

Categorical (Qualitative)

- Nominal scales – number is just a symbol that identifies a quality
  - 0=male, 1=female
  - 1=green, 2=blue, 3=red, 4=white
- Ordinal – rank order

Quantitative (continuous and discrete)

- Interval – units are of identical size (i.e. Years)
- Ratio – distance from an absolute zero (i.e. Age, reaction time)

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What is a measurement?

Every measurement has 2 parts:

The True Score (the actual state of things in the world)

and

ERROR! (mistakes, bad measurement, report bias, context effects, etc.)

$$X = T + e$$

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Organizing your data in a spreadsheet

Stacked data:

- Multiple cases (rows) for each subject

Subject	condition	score
1	before	3
1	during	2
1	after	5
2	before	3
2	during	8
2	after	4
3	before	3
3	during	7
3	after	1

Unstacked data:

- Only one case (row) per subject

Subject	before	during	after
1	3	2	5
2	3	8	4
3	3	7	1

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Handling missing data

- **Removing missing data**

Once you know that data is MCAR and a relatively small fraction of observations have missing values, then it may be safe to remove observations

- **Encoding as missing**

In the MCAR or MAR case for categorical attributes y, a useful approach is to encode the fact that a value is missing as a new category and include that in subsequent analysis of attribute y.

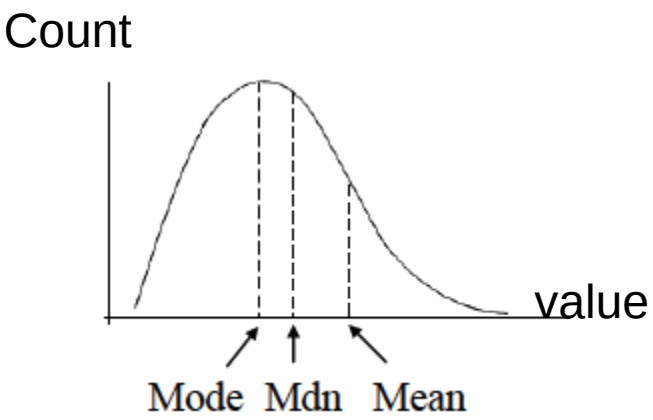
- **Imputation**
  - **MCAR (also for MAR, but this is not ideal),** In this case we can use a simple method for imputation of y. For numeric attributes we replace missing values in y with the mean of non-missing values of y.
  - **Predicting model for MAR, i**n this case we use a more complex method by replacing missing values for attribute y predicting from other variables x when variables are related (e.g. Linear regression)

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Variable Summaries

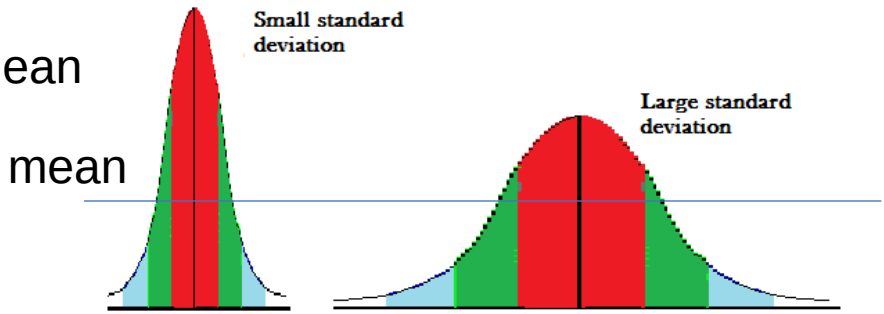
Indices of central tendency:

- Mean – the average value
- Median – the middle value
- Mode – the most frequent value



Indices of Variability:

- Variance – the spread around the mean
- Standard deviation
- Standard error of the mean (estimate)



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The Mean

Subject	before	during	after
1	3	2	7
2	3	8	4
3	3	7	3
4	3	2	6
5	3	8	4
6	3	1	6
7	3	9	3
8	3	3	6
9	3	9	4
Sum =	30	50	50
/n	10	10	10
Mean =	3	5	5

Mean = sum of all scores divided by number of scores

$$\frac{X_1 + X_2 + X_3 + \ldots X_n}{n}$$

mean and median applet

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The Variance: Sum of the squared deviations divided by number of scores

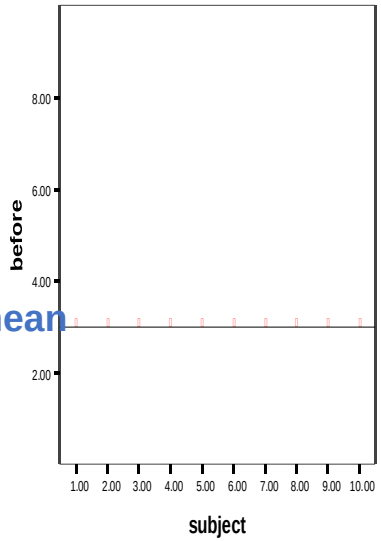
Subject	before	during	after
1	3	2	7
2	3	8	4
3	3	7	3
4	3	2	6
5	3	8	4
6	3	1	6
7	3	9	3
8	3	3	6
9	3	9	4
10	3	1	5
Sum =	30	50	50
N	10	10	10
Mean =	3	5	5

Before –mean	Before – mean^2	During – mean	During – mean^2	After - mean	After – mean^2
0	0	-3	9	2	4
0	0	3	9	-1	1
0	0	2	4	-2	4
0	0	-3	9	1	1
0	0	3	9	-1	1
0	0	-4	16	1	1
0	0	4	16	-2	4
0	0	-2	4	1	1
0	0	4	16	-1	1
0	0	-4	16	2	4
0	0	0	108	0	22
	10*		10		10
VAR =	0		10.8		2.2

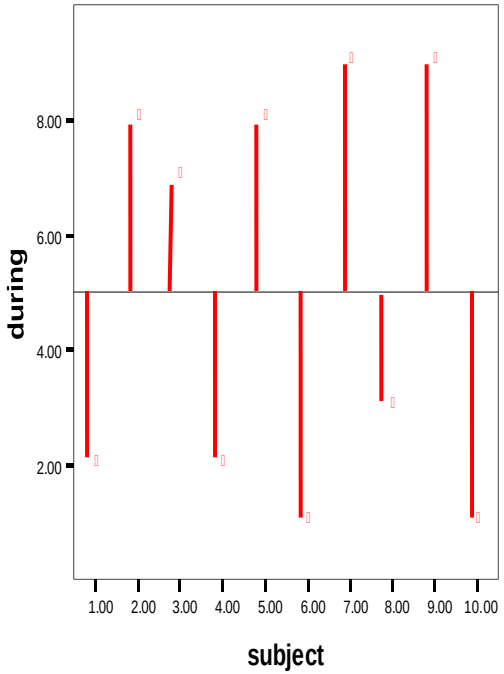
*actually you divide by n-1 because it is a sample and not a population, but you get the idea…

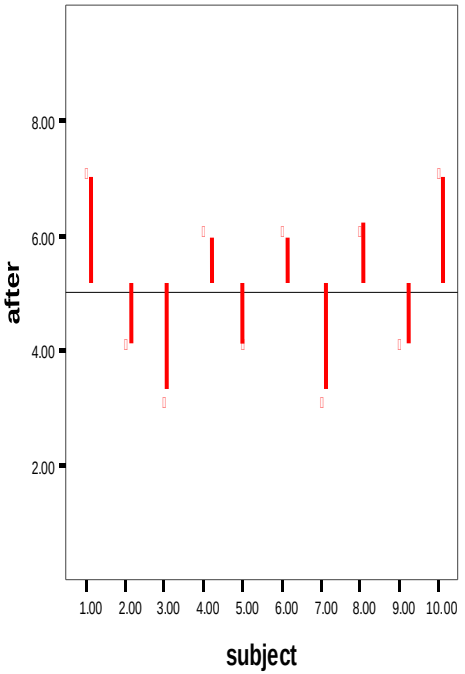
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Variance continued



mean





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Standard Variance calculation

The process of standardizing deviations goes like this:

1. Score (in the units that are meaningful)
2. Mean
3. Each score's deviation from the mean
4. Square that deviation
5. Sum all the squared deviations (Sum of Squares)
6. Divide by n (if population) or n-1 (if sample)
7. Square root – now the value is in the units we started with!!!

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Standardize Variance

Variance, as calculated earlier, is arbitrary.

What does it mean to have a variance of 10.8? Or 2.2? Or 1459.092? Or 0.000001?

Nothing. But if you could “standardize” that value, you could talk about any variance (i.e. deviation) in equivalent terms.

Standard Deviations are simply the square root of the variance

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Distribution

Means and variances are ways to describe a *distribution* of scores/values.

Knowing about your distributions is one of the best ways to understand your data

A NORMAL (aka Gaussian) distribution is the most common assumption of statistics, thus it is often important to check if your data are normally distributed.

We will discuss more Gaussian in ML part.

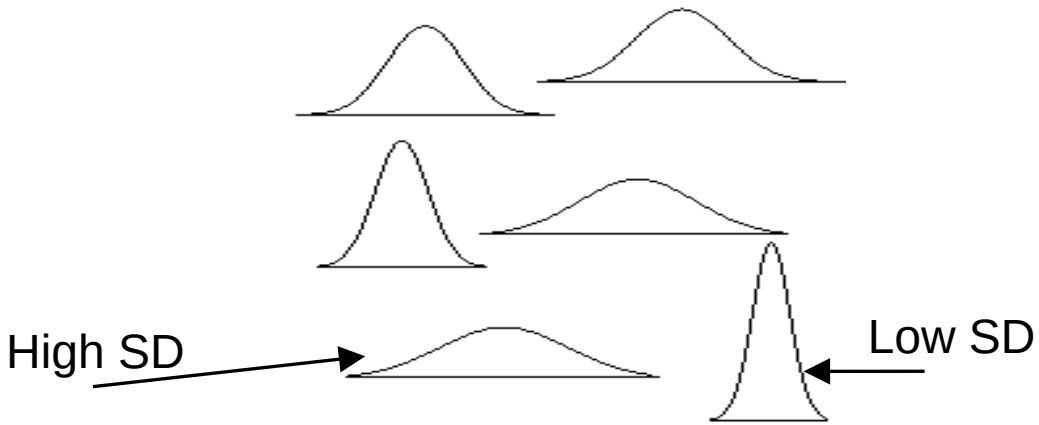
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Interpreting standard deviation (SD)

First, the SD will let you know about the distribution of scores around the mean.

High SDs (relative to the mean) indicate the scores are spread out

Low SDs tell you that most scores are very near the mean.



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Standardized scores (Z)

Then, you can use SDs to create *standardized* scores – that is, force the scores onto a normal distribution by putting each score into units of SD.

Subtract the mean from each score and divide by SD

$$Z = (X - \text{mean})/SD$$

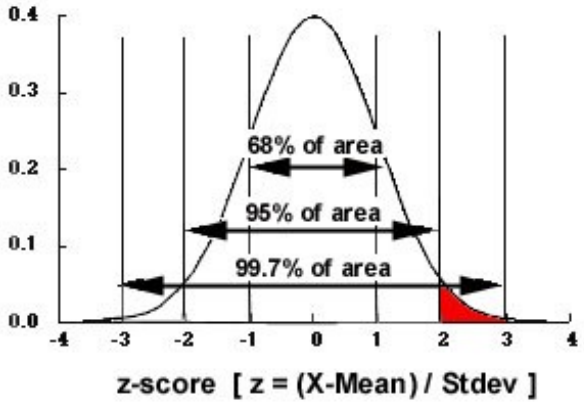
This is truly an amazing thing

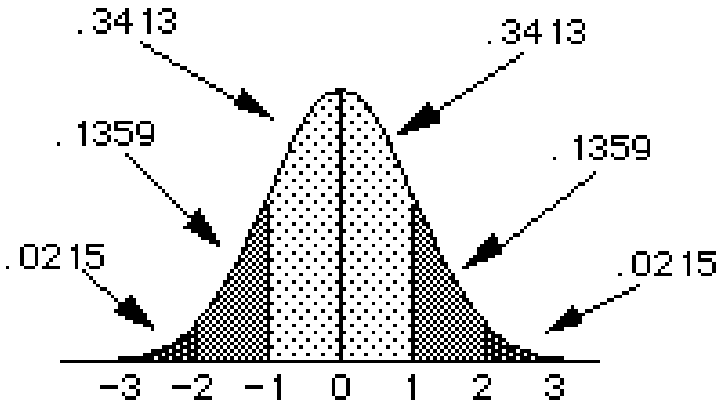
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Standardized normal distribution

ALL Z-scores have a mean of 0 and SD of 1. Nice and simple.

From this we can get the proportion of scores anywhere in the distribution.





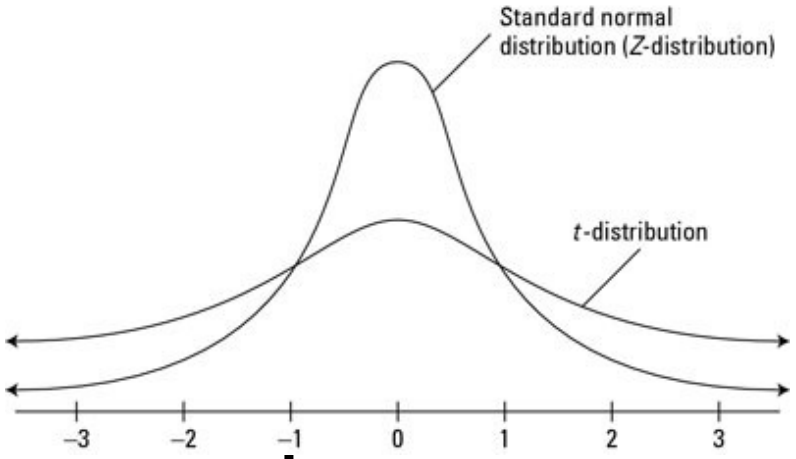
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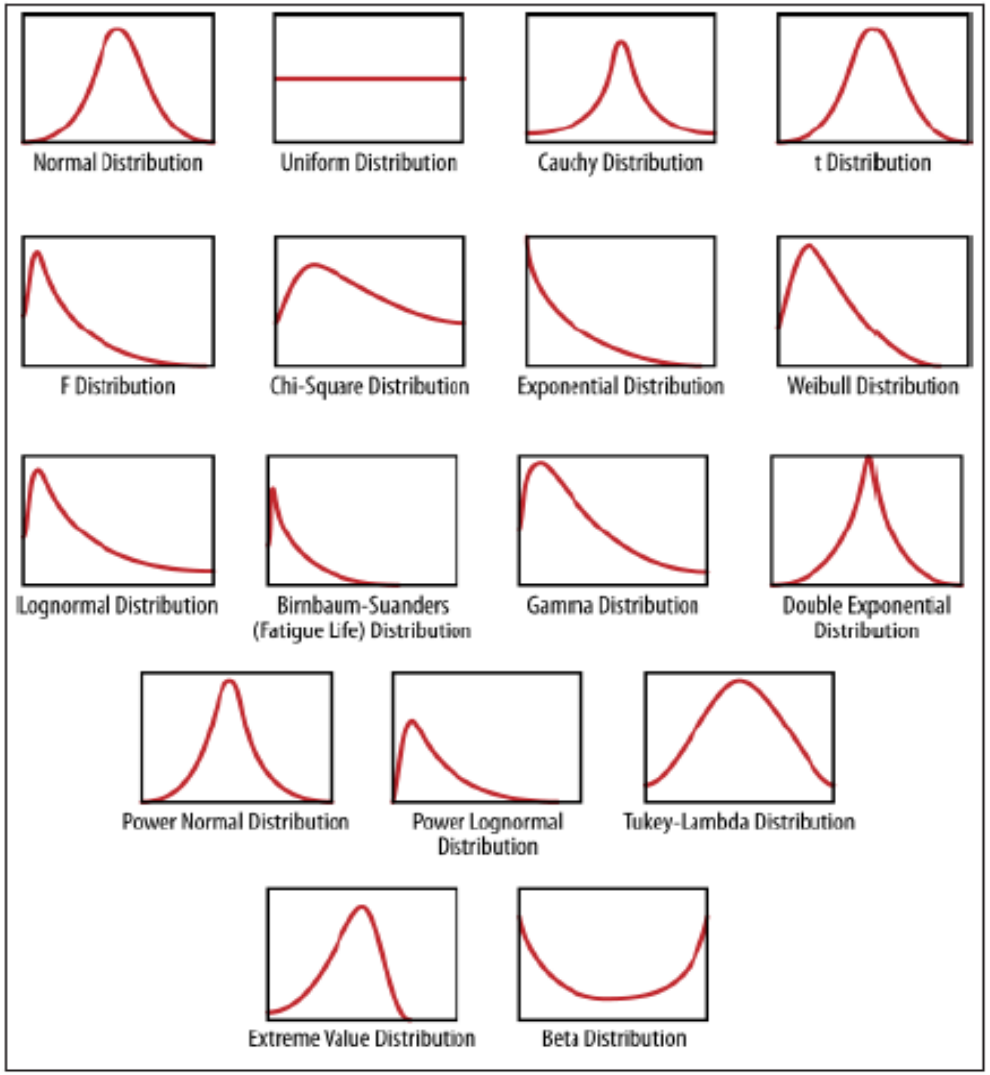
The trouble with normal

- We violate assumptions about statistical tests if the distributions of our variables are not approximately normal.
- Thus, we must first examine each variable's distribution and make adjustments when necessary so that assumptions are met.
- More types of distribution need to be considered.

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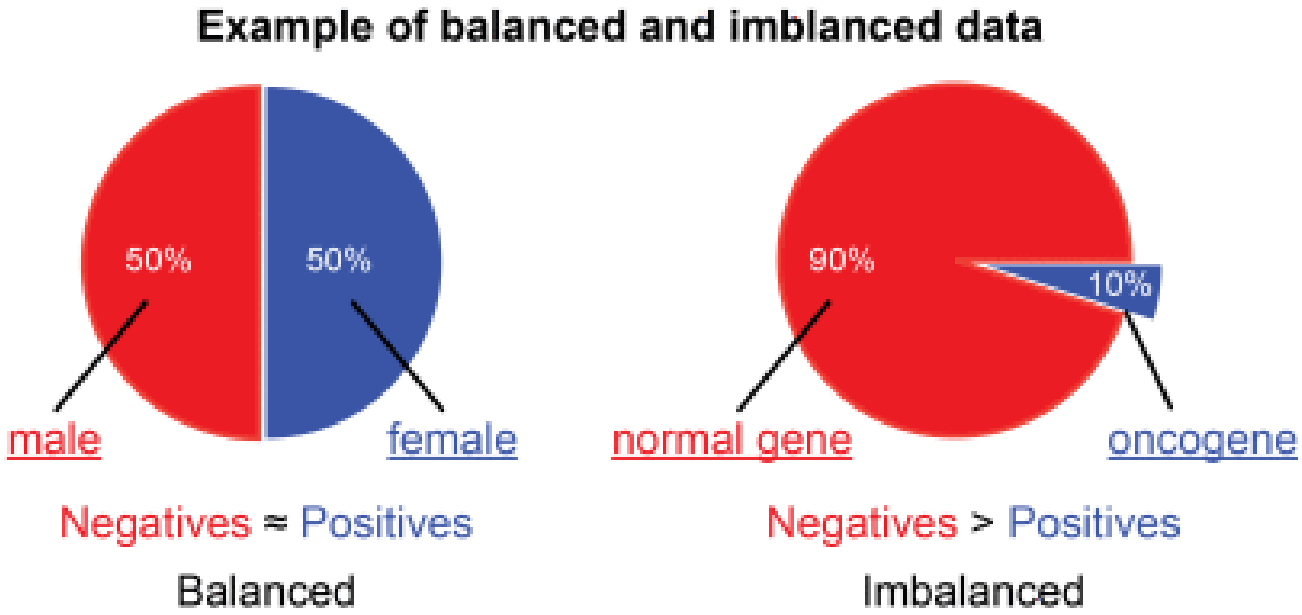
A bunch of continuous probability distribution





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Checking data balance



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Visual display of univariate data

- Now the example data from before has decimals

  (what kind of data is that?)

- Precision has increased

Subject	before	during	after
1	3.1	2.3	7
2	3.2	8.8	4.2
3	2.8	7.1	3.2
4	3.3	2.3	6.7
5	3.3	8.6	4.5
6	3.3	1.5	6.6
7	2.8	9.1	3.4
8	3	3.3	6.5
9	3.1	9.5	4.1
10	3	1	7.3

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Visual display of univariate data (Cont.)

- Histograms
- Stem and Leaf plots
- Boxplots
- QQ Plots

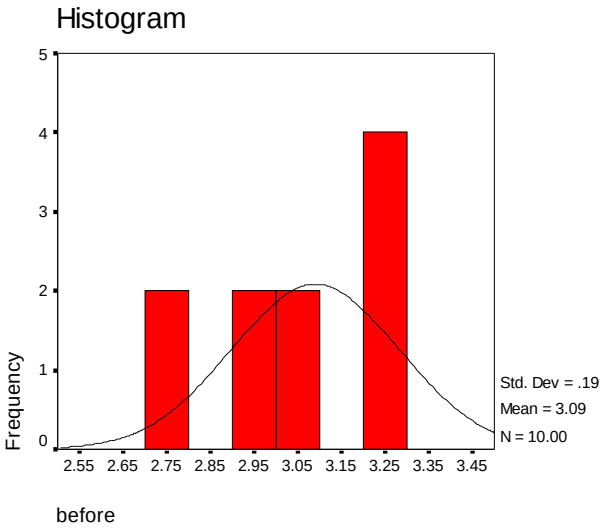
…and many many more

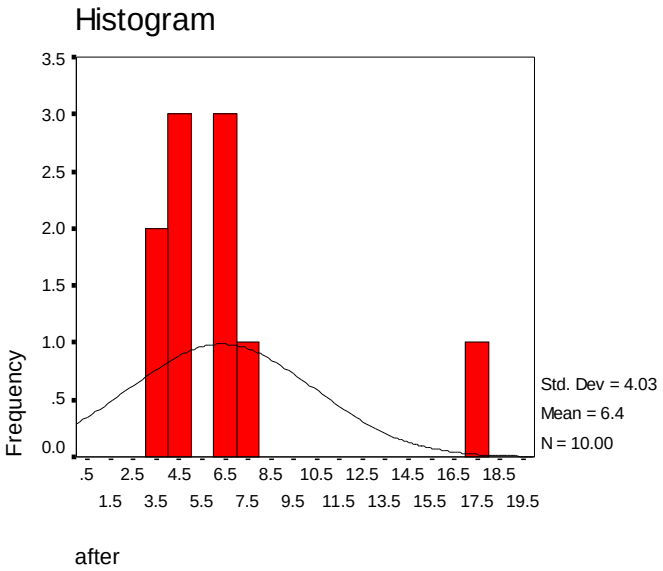
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2	3.2	8.8	4.2
3	2.8	7.1	3.2
4	3.3	2.3	6.7
5	3.3	8.6	4.5
6	3.3	1.5	6.6
7	2.8	9.1	3.4
8	3	3.3	6.5
9	3.1	9.5	4.1
10	3	1	7.3

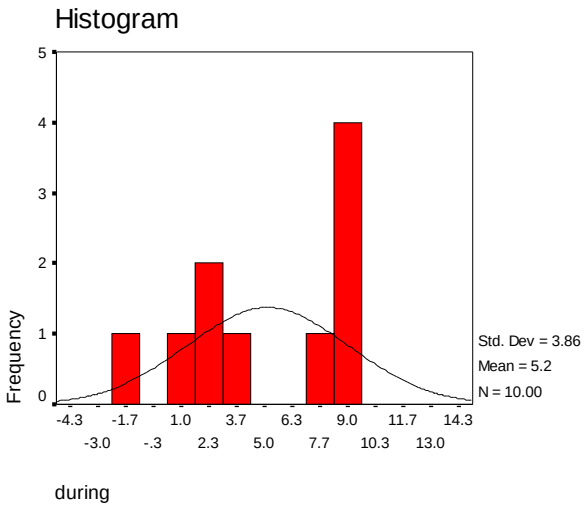
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Histograms

- # of bins is very important







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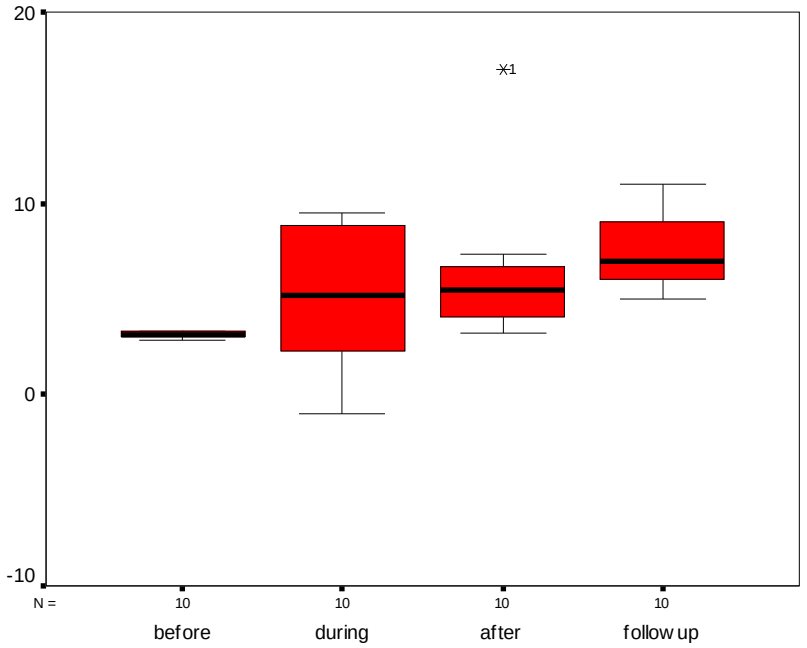
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Boxplots

Upper and lower bounds of boxes are the 25th and 75th percentile (interquartile range)

Whiskers are min and max value unless there is an outlier

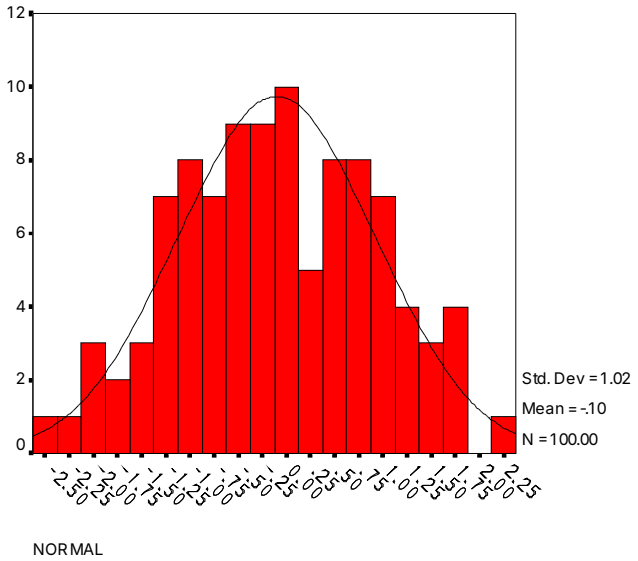
An outlier is beyond 1.5 times the interquartile range (box length)



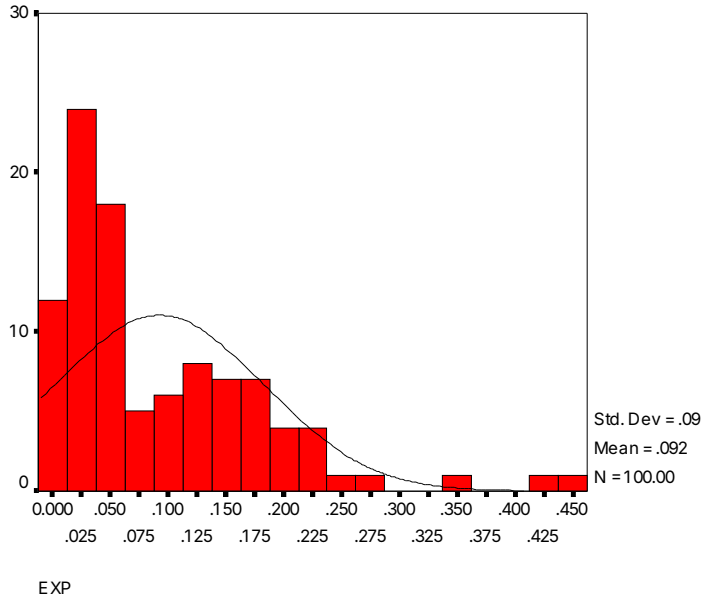
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Quantile-Quantile (Q-Q) Plots

The quantile-quantile (q-q) plot is a graphical technique for determining if two data features or sets come from populations with a common distribution.



Random Normal Distribution



Random Exponential Distribution

Vertical axis: Estimated quantiles from data feature 1
Horizontal axis: Estimated quantiles from data feature 2

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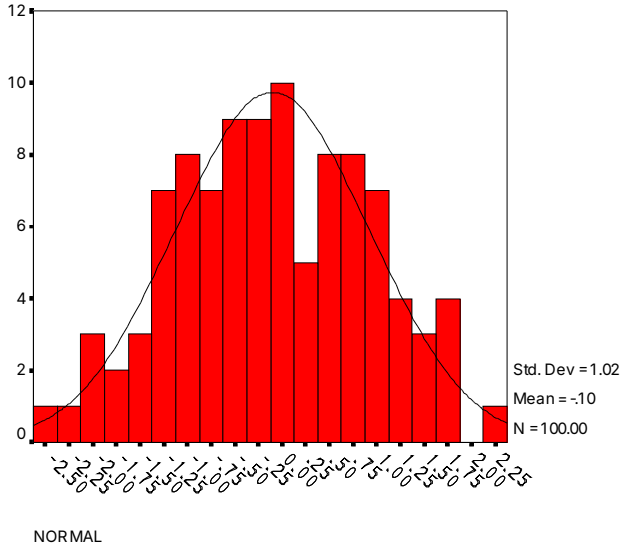
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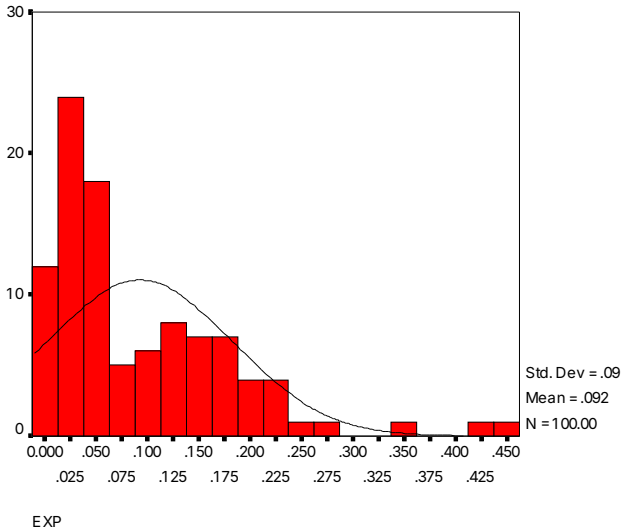
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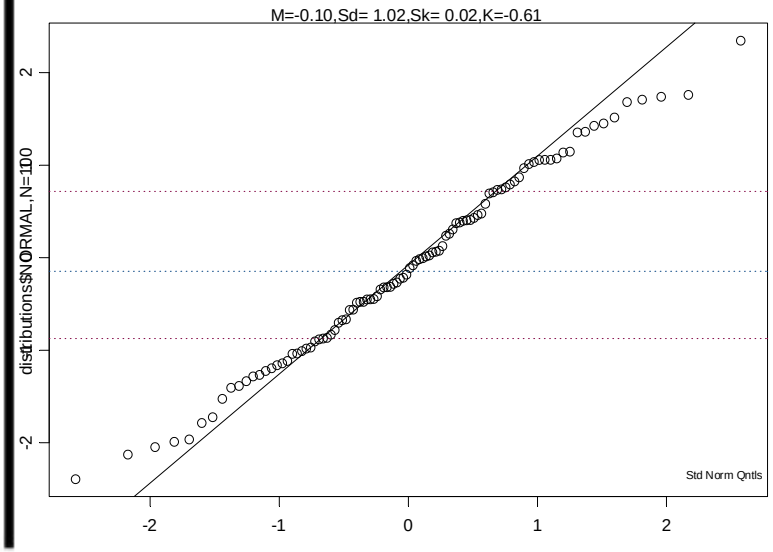
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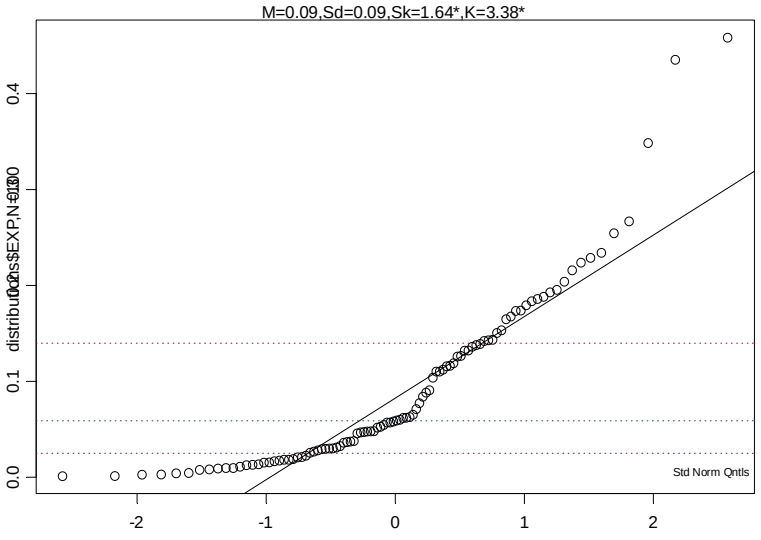
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Q-Q Plots









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So…what do you do with EDA?

If you find a mistake, fix it.

If you find an outlier, trim it or delete it.

If your distributions are a skew, transform the data.

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Dealing with Outliers

First, try to explain it.

In a normal distribution 0.4% are outliers (>2.7 SD) and 1 in a million is an extreme outlier (>4.72 SD).

For analyses you can:

Delete the value – crude but effective

Change the outlier to value ~3 SD from mean

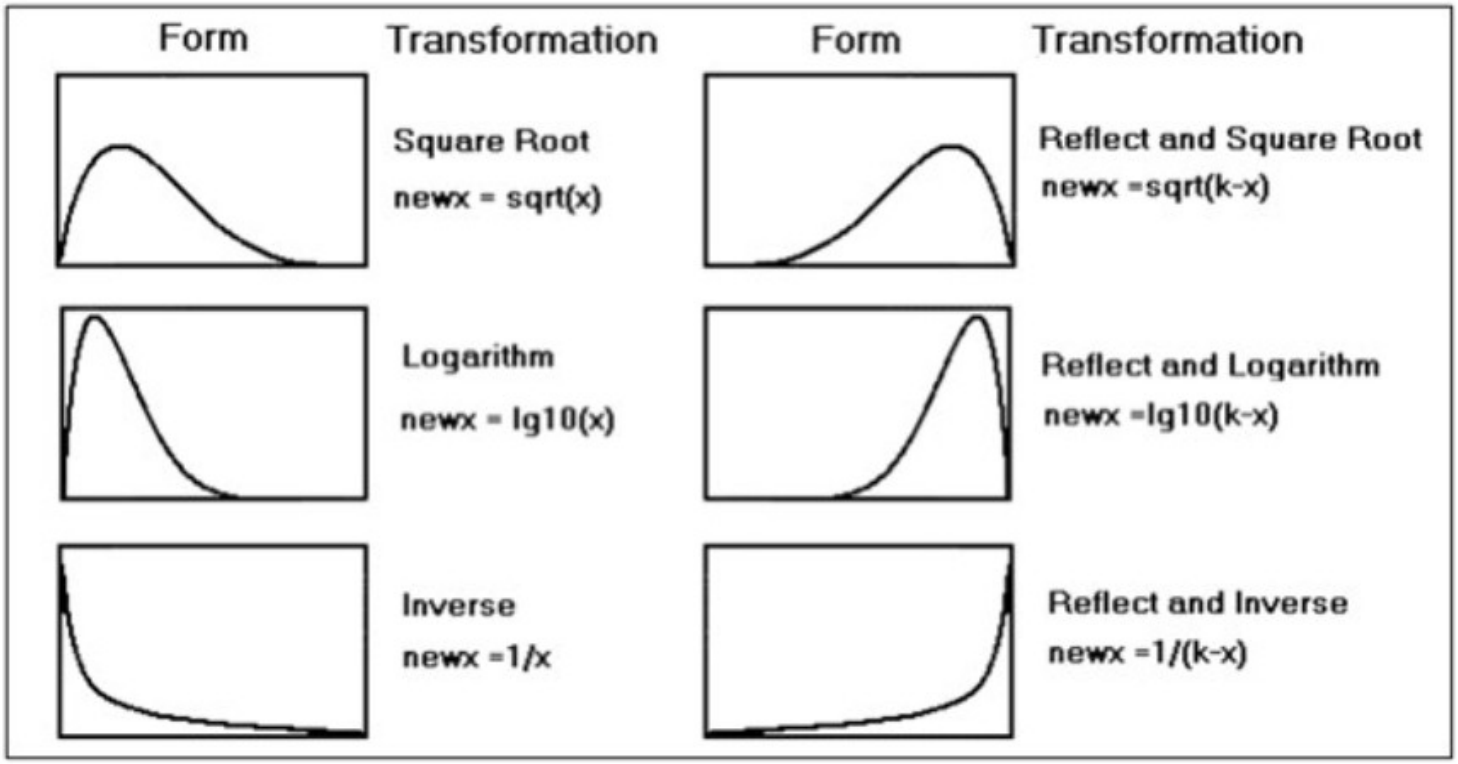
“Winsorize” it (make = to next highest value)

“Trim” the mean – recalculate mean from data within interquartile range

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Dealing with skewed distributions

(Skewness and kurtosis greater than +/- 2)

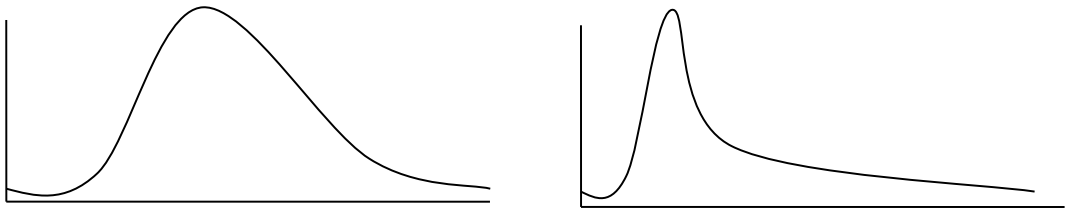


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Visual Display of Bivariate Data

So, you have examined each variable for mistakes, outliers and distribution and made any necessary alterations. Now what?

Look at the relationship between 2 (or more) variables at a time



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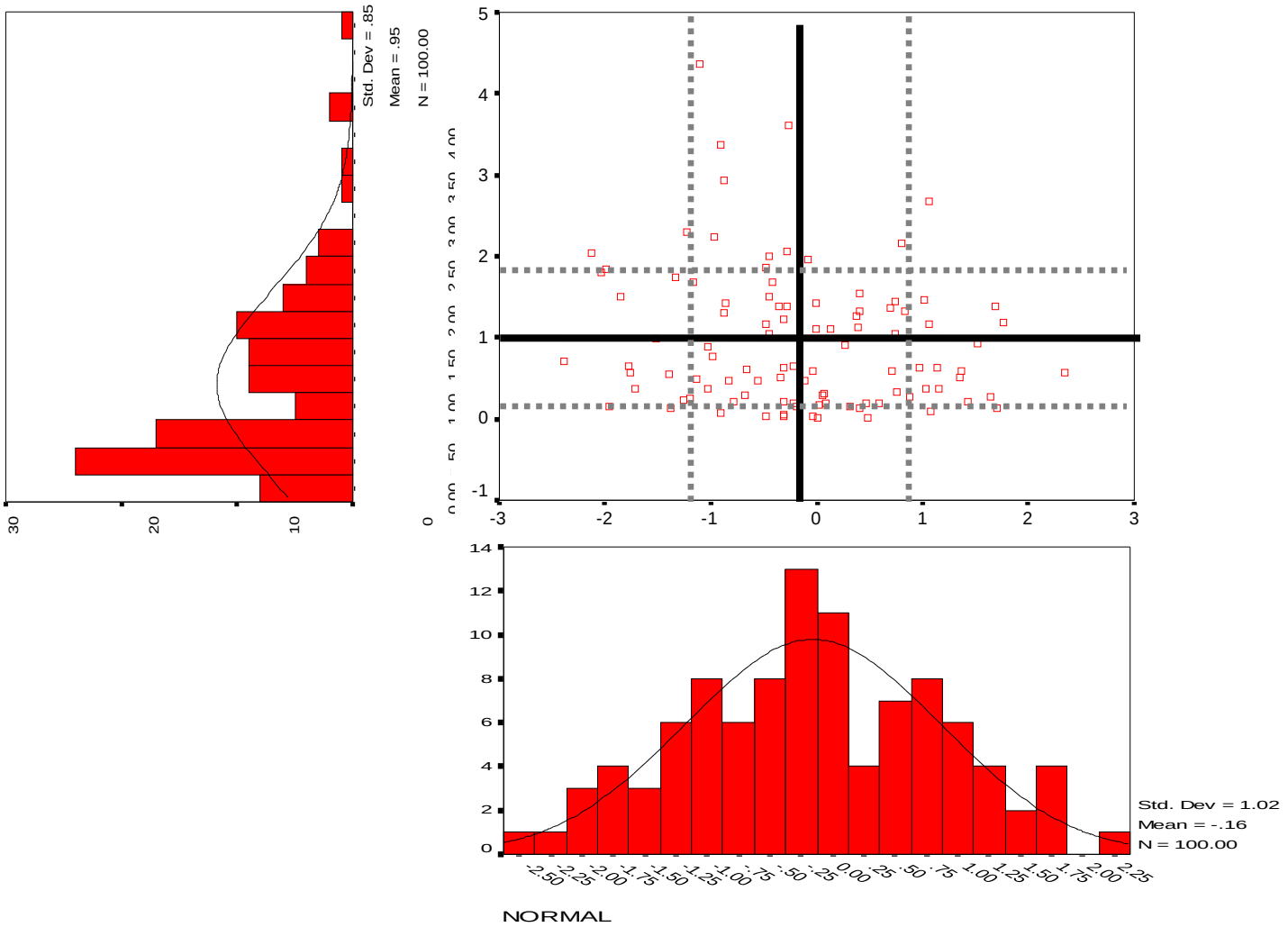
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Visual Displays of Bivariate Data

Variable 1	Variable 2	Display Example
Categorical	Categorical	Crosstabs
Categorical	Continuous	Box plots
Continuous	Continuous	Scatter plots

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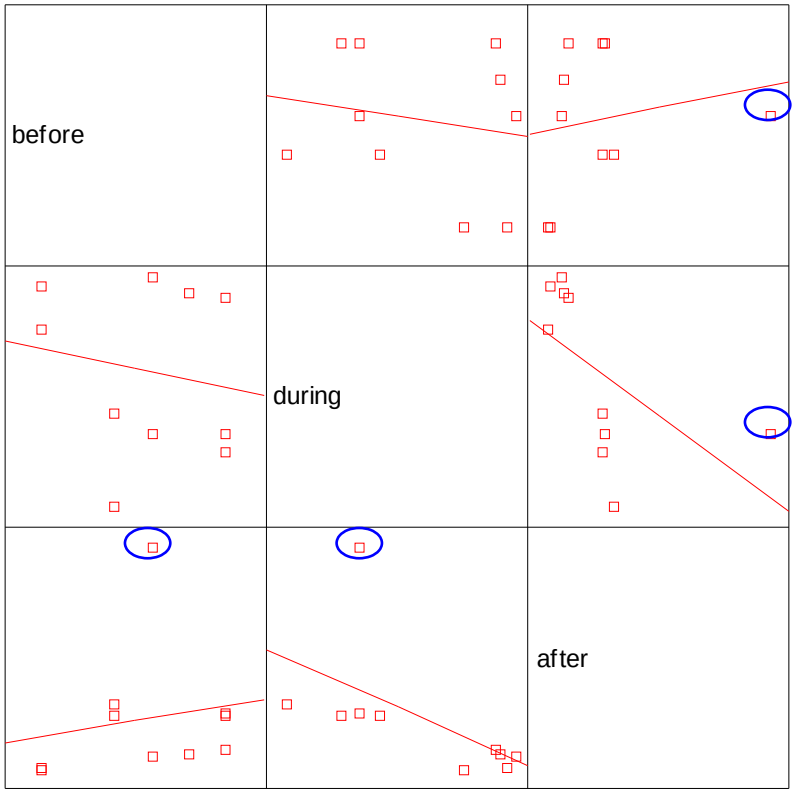
Bivariate Distribution



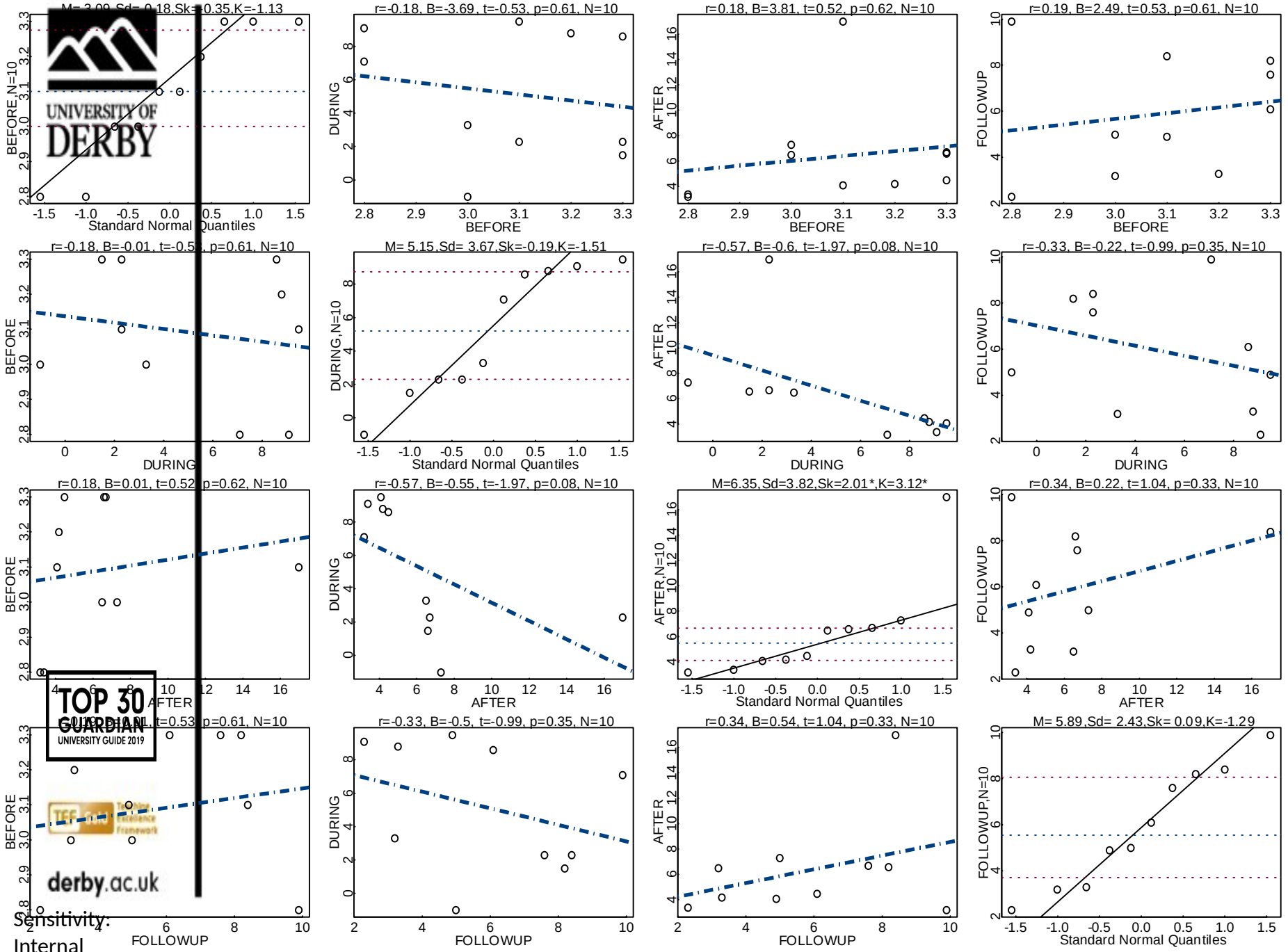
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Intro to Scatter plots

Scatter plots uses dots to represent values for two different numeric variables. The position of each dot on the horizontal and vertical axis indicates values for an individual data point. **Scatter plots** are used to observe relationships between variables.



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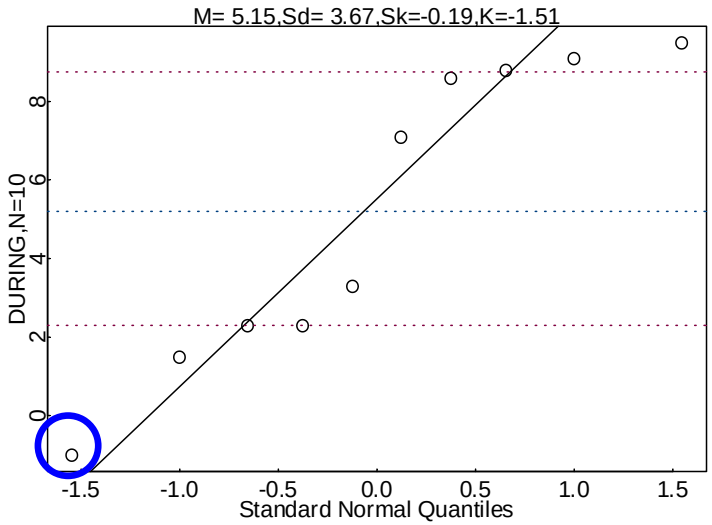
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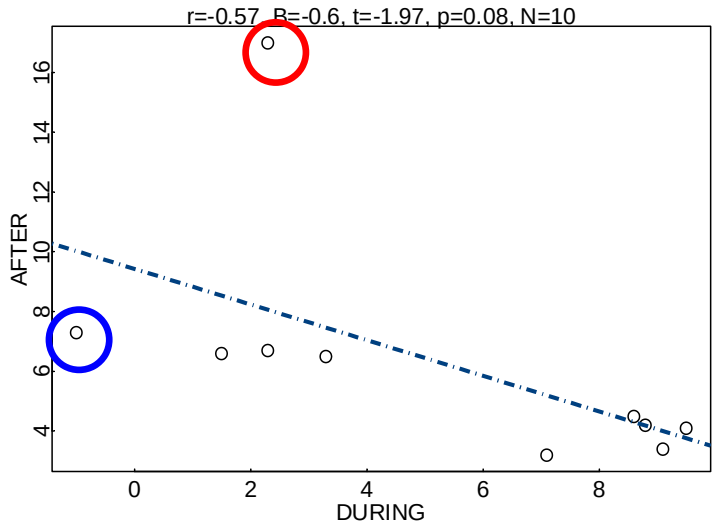
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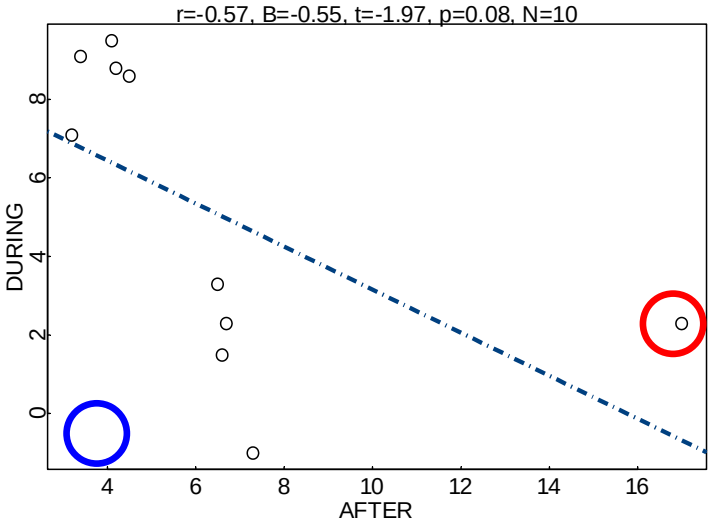
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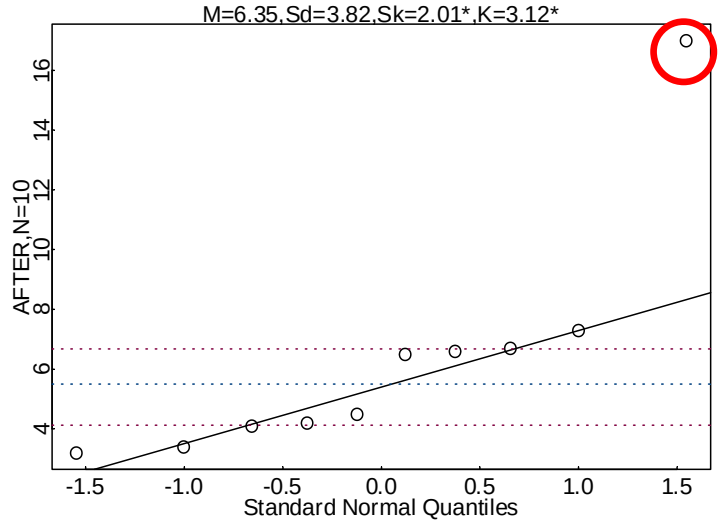
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With Outlier and Out of Range Value



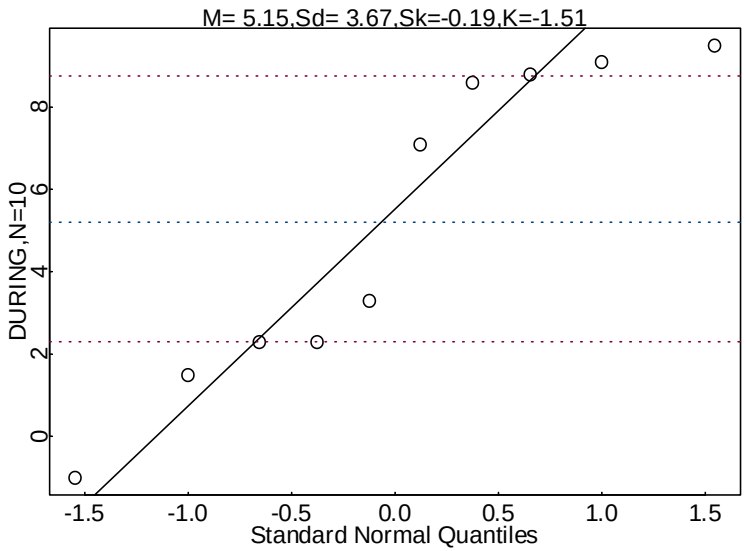


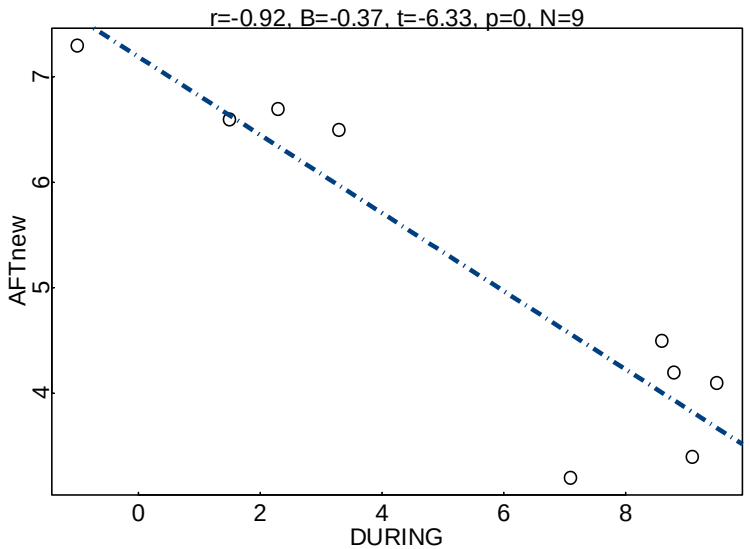


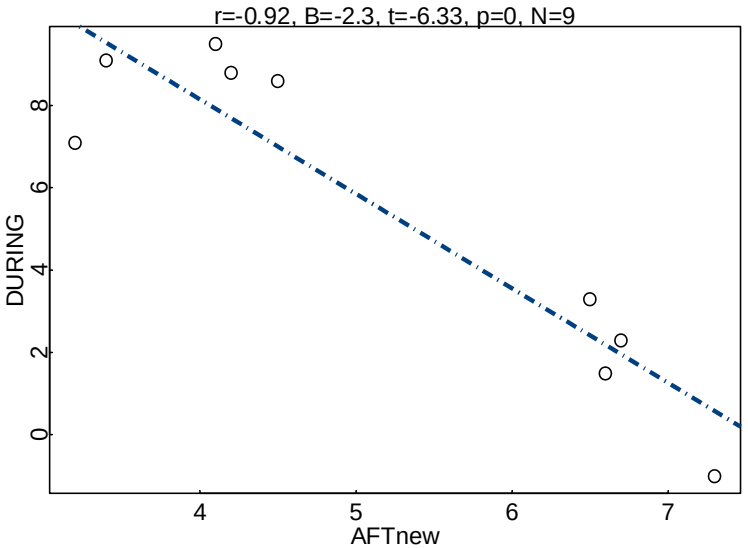


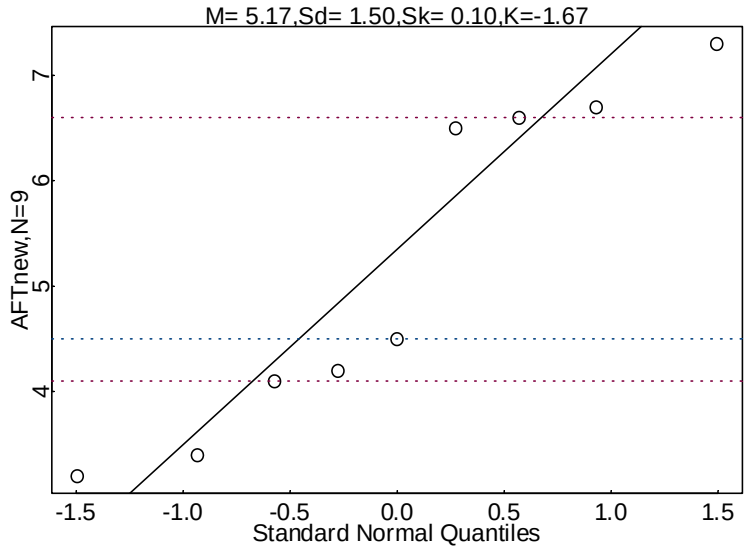
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Without Outlier



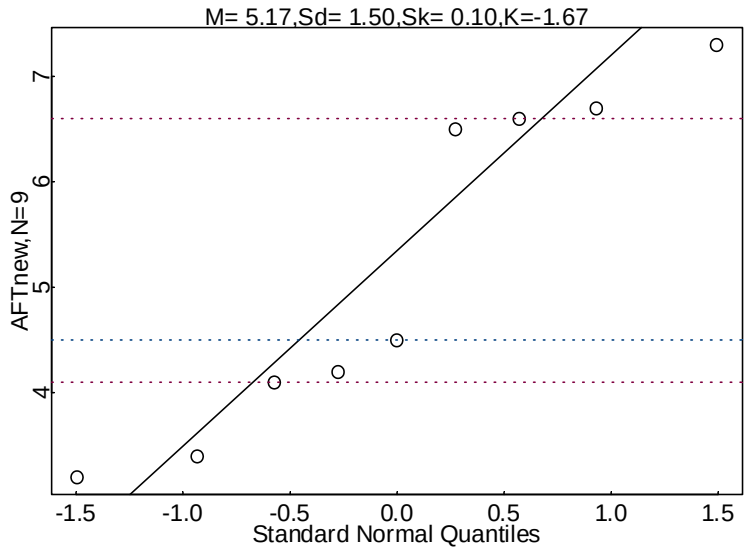


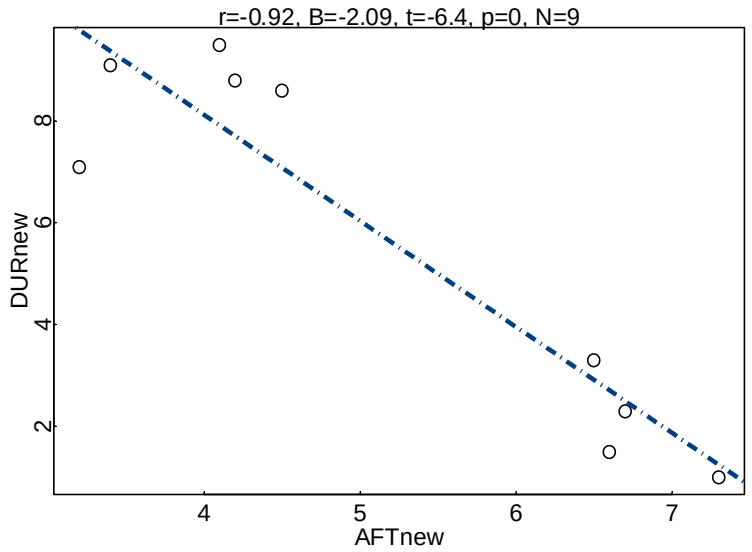


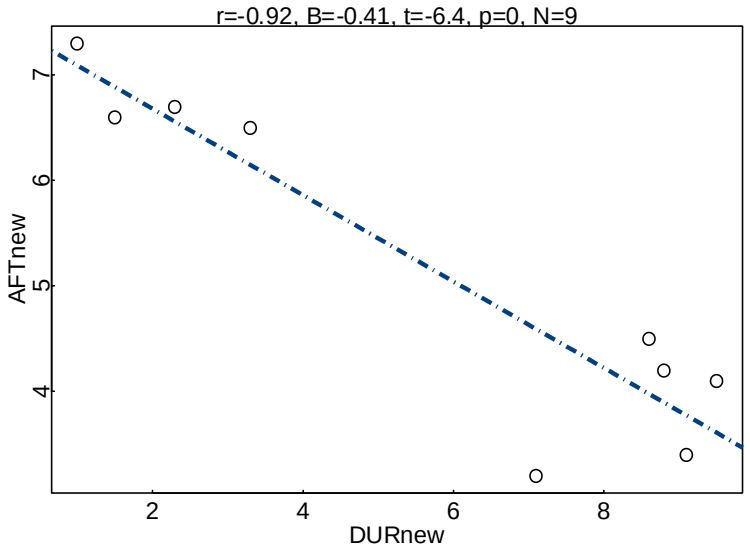


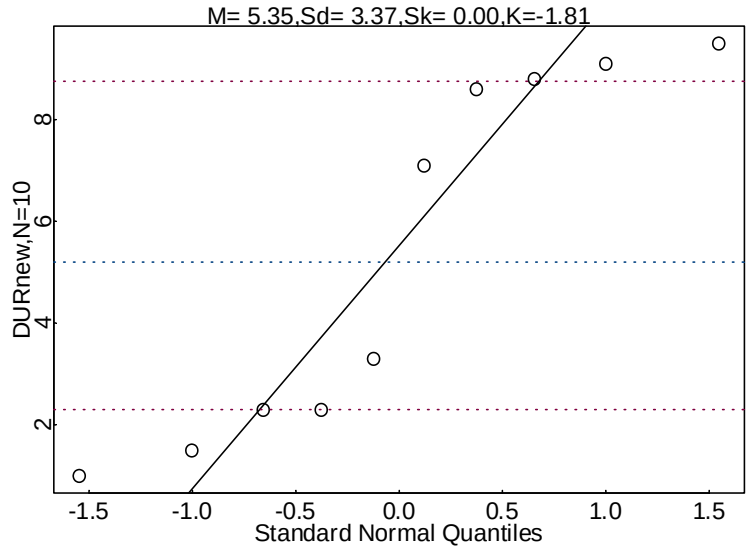
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With Corrected Out of Range Value









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Scales of Graphs

It is very important to pay attention to the *scale* that you are using when you are plotting.

Compare the following graphs created from identical data

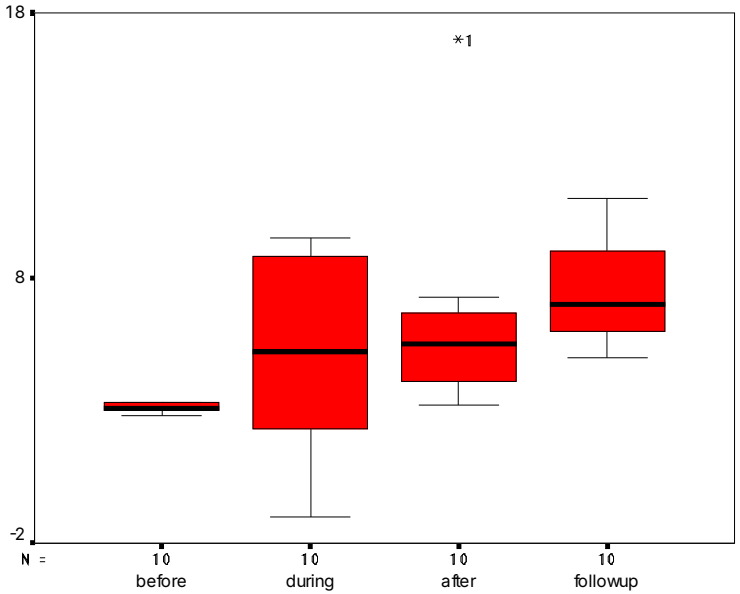
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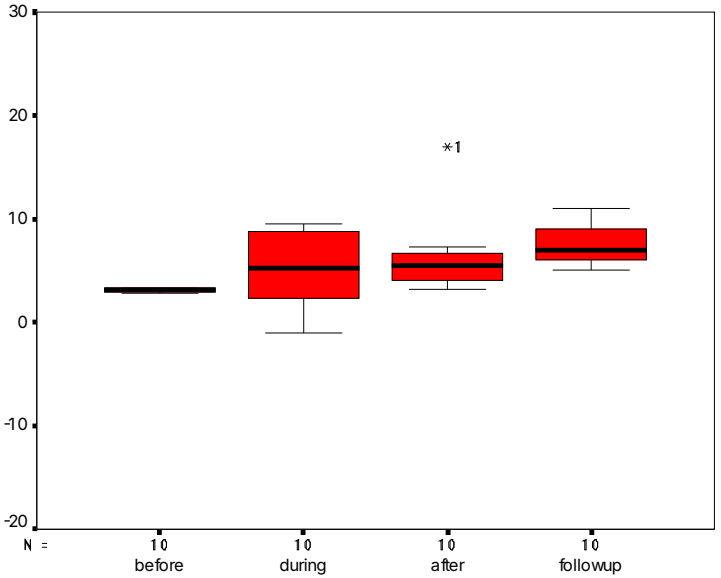
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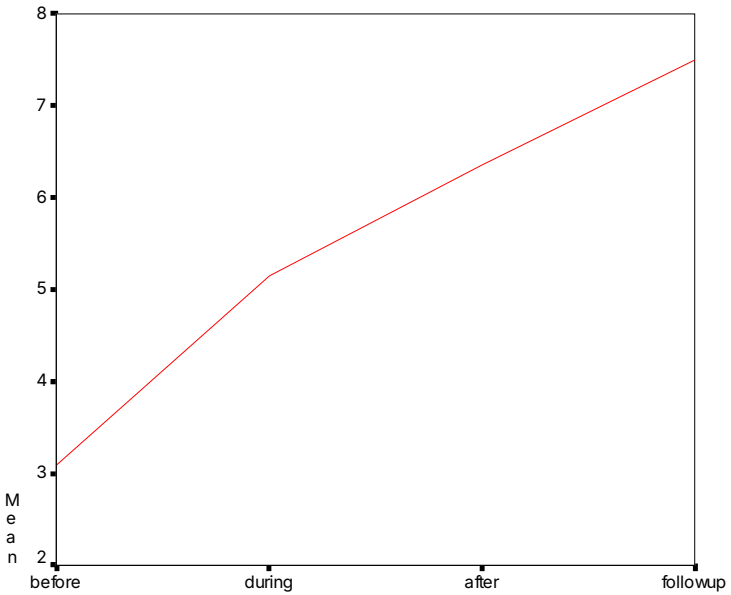
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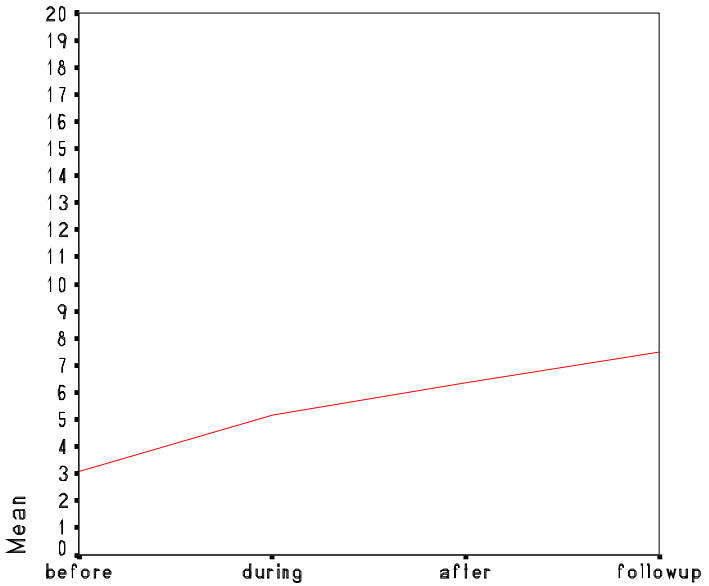
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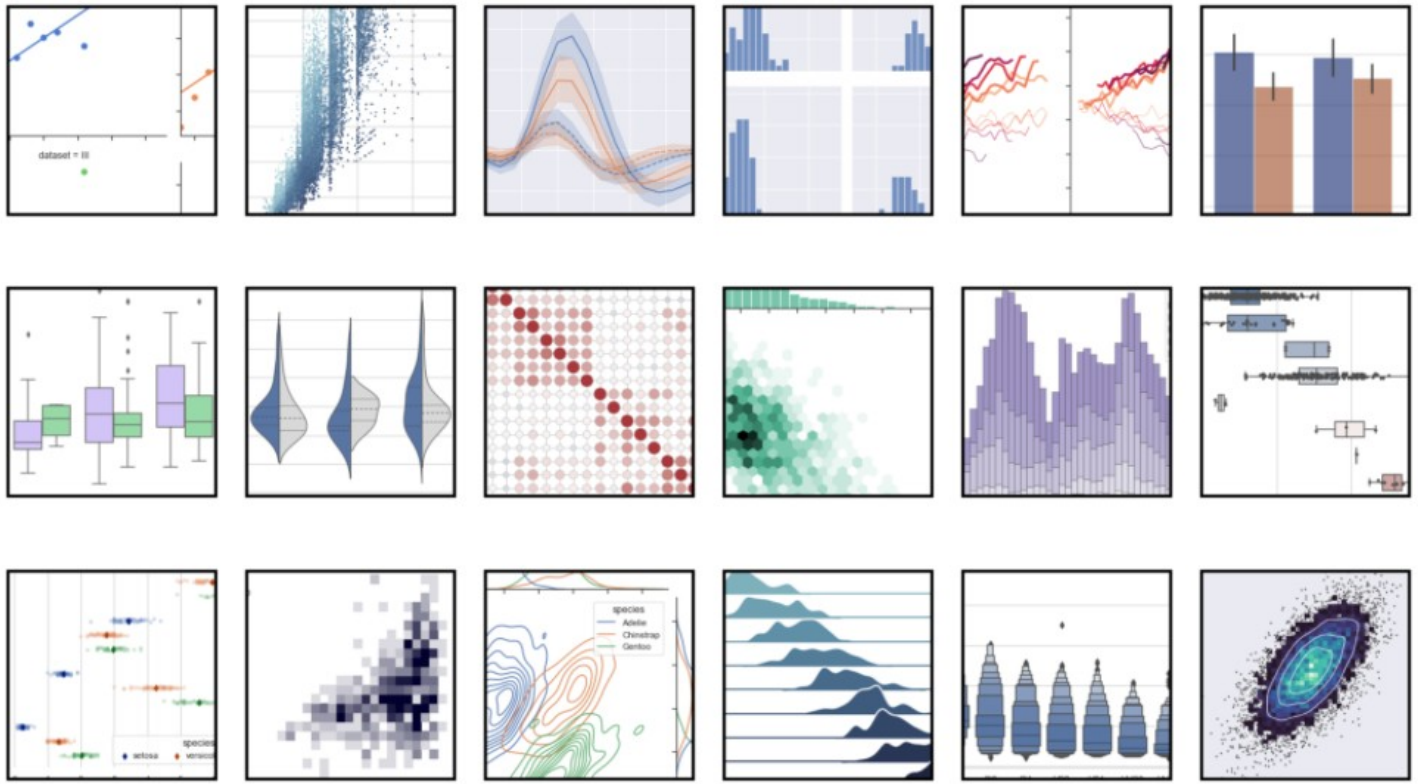
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Python visualisation tools

- Matplotlib (https://matplotlib.org/)
- Seaborn (https://seaborn.pydata.org/)

Example gallery



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Summary

Examine all your variables thoroughly and carefully before you begin analysis

Use visual displays whenever possible

Transform each variable as necessary to deal with mistakes, outliers, and distributions

Let's do practices on Kaggle

Homework: read Chapter 2 of Doing Data Science

Next week:

Lecture: start mining the data for frequent patterns

Practices: Association rule mining for frequent patterns

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More references

1. *Exploratory Data Analysis* by John Tukey (Pearson)

2. *The Visual Display of Quantitative Information* by Edward Tufte (Graphics Press)

3. *The Elements of Graphing Data* by William S. Cleveland (Hobart Press)

4. *Statistical Graphics for Visualizing Multivariate Data* by William G. Jacoby (Sage)

5. *"Exploratory Data Analysis for Complex Models"* by Andrew Gelman (American Statistical Association)

6. The Future of Data Analysis (PDF) by John Tukey. Annals of Mathematical Statistics, Volume 33, Number 1 (1962), 1-67.

7. *Data Analysis, Exploratory* by David Brillinger [8-page excerpt from International *Encyclopedia of Political Science* (Sage)]

8. https://www.hcbravo.org/IntroDataSci/bookdown-notes/eda-handling-missing-data.html

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